CloudXeus

AI-103 Capstone - Advanced Parsing Demo

Synthetic scanned PDF with a multi-page table for OCR and structure-aware ingestion

Section 1 - Course Delivery and Security Setup

Document ID: AI103-RAG-ADV-PARSING-DEMO

Purpose: Use this file to show why advanced data parsing is needed for scanned PDFs with tables that continue across multiple pages.

Parsing target: OCR + table structure + headings + page-number metadata

Table 1 - CloudXeus Capstone Lab and Evidence Matrix

Step	Section	Requirement / Lab task	Azure service	Expected evidence	Pg
1	Course overview	Identify the goal of the AI-103 capstone and explain how CloudXeus uses an agent to answer course product questions.	Microsoft Foundry, Azure AI Search	Student can describe the business scenario and end-to-end architecture.	1
2	Data source	Upload five product sheet PDFs for AZ-104, AI-901, AI-103, DP-600, and DP-700 into the course-products container.	Azure Blob Storage	Blob folder structure is visible and each PDF is stored under the correct course folder.	1
3	Data source	Confirm that product sheets contain headings, tables, and visual sections so retrieval can be tested against mixed content.	Azure Blob Storage	Documents include structured sections and embedded visuals for extraction testing.	1
4	Security	Enable a system-assigned managed identity for the Azure AI Search service.	Azure AI Search	Search service identity is visible in the Identity blade.	1
5	Security	Grant the Search service managed identity Storage Blob Data Reader on the storage account.	Azure Storage RBAC	Search can read source PDFs without using account keys in the pipeline.	1
6	Security	Grant the Search service managed identity Cognitive Services User on the Foundry resource.	Microsoft Foundry RBAC	Search can access AI services used during ingestion and retrieval.	1
7	Security	Grant the Search service managed identity Cognitive Services OpenAI User on the Foundry resource.	Microsoft Foundry RBAC	Search can call embedding and chat completion deployments for vectorization and planning.	1
8	Knowledge source	Create an Azure Blob indexed knowledge source named ks-cloudxeus-course-products-blob.	Azure AI Search	Generated datasource, skillset, indexer, and index are created for the Blob container.	1
9	Parsing	Choose Standard content extraction for product PDFs with tables, images, headings, and course diagrams.	Azure AI Search	Indexer pipeline performs richer extraction compared to plain text extraction.	1

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Section 1 continued - Knowledge Source, Vectorization, and Indexing

Table 1 - CloudXeus Capstone Lab and Evidence Matrix (continued)

Rows continue from the previous page. Header row is repeated to preserve column meaning.

Step	Section	Requirement / Lab task	Azure service	Expected evidence	Pg
10	Vectorization	Add a vectorizer using the text-embedding-3-large deployment.	Azure OpenAI / Foundry	Document chunks are converted into vectors for semantic similarity retrieval.	2
11	Image handling	Enable image verbalization for PDFs that contain diagrams and embedded visual sections.	Azure AI Search Skillset	Images are represented as descriptive text that can be retrieved later.	2
12	Knowledge base	Create kb-cloudxeus-course-products and attach the Blob knowledge source.	Azure AI Search	Knowledge base becomes the retrieval layer for agentic retrieval.	2
13	Retrieval	Set reasoning effort to Low for the first demo run to reduce cost and latency while testing retrieval quality.	Azure AI Search	Knowledge base is ready for basic query planning against product sheets.	2
14	Testing	Ask: Which CloudXeus course prepares students for developing AI apps and agents on Azure?	Knowledge base chat	Expected answer points to AI-103 and cites course product evidence.	2
15	Testing	Ask: Compare AI-901 and AI-103 for a student who wants to build Azure AI solutions.	Agentic retrieval	System retrieves evidence from multiple course PDFs before answering.	2
16	Testing	Ask a follow-up: Does this course include hands-on labs?	Agentic retrieval	Conversation context influences retrieval planning for the follow-up question.	2
17	Troubleshooting	If the indexer fails with Unauthorized, inspect whether the Search managed identity has both Foundry roles.	Azure RBAC	Indexer errors are mapped to missing model access or missing storage access.	2
18	Troubleshooting	If the indexer fails before processing documents, verify Storage Blob Data Reader and storage networking settings.	Azure Storage	Source document access is restored and indexer can enumerate blobs.	2
19	Indexer	Reset and rerun the generated indexer after fixing role assignments.	Azure AI Search	All five documents process successfully after permissions propagate.	2
20	Indexer	Inspect the generated index to confirm chunk, vector, metadata, and source fields were created.	Azure AI Search	Students understand how the knowledge source created supporting search objects.	2

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Section 2 - Advanced Parsing, Evaluations, and Instructions

Table 1 - CloudXeus Capstone Lab and Evidence Matrix (continued)

Rows continue from the previous page. Header row is repeated to preserve column meaning.

Step	Section	Requirement / Lab task	Azure service	Expected evidence	Pg
21	Advanced parsing	Reingest a scanned PDF that contains this table spanning multiple pages.	Azure AI Search	OCR is required because the PDF pages are image-based and not text-selectable.	3
22	Advanced parsing	Preserve table headings across page boundaries so each chunk can keep its column meaning.	Advanced data parsing	Rows on continuation pages retain the context of Step, Section, Task, Service, Evidence, and Page.	3
23	Advanced parsing	Preserve page-number metadata with every table chunk.	Advanced data parsing	Retrieved evidence can identify the page on which a row was found.	3
24	Advanced parsing	Avoid storing each scanned page as one large chunk when users ask for specific rows.	Advanced data parsing	Retrieval can target precise table rows instead of returning an entire page.	3
25	Advanced parsing	Avoid basic fixed-size chunking when the document contains headings and multipage tables.	Advanced data parsing	Chunks remain structure-aware and do not split rows or lose heading context.	3
26	Evaluation	Create test questions for each course and compare answers against the product sheets.	Foundry evaluations	Evaluation data helps measure relevance, groundedness, and answer completeness.	3
27	Evaluation	Add negative tests, such as asking about unrelated AWS discount policies or personal student data.	Foundry evaluations	Agent should refuse or ask for clarification when source data is missing.	3
28	Instructions	Add system instructions that restrict the agent to CloudXeus product information.	Foundry Agent	Agent avoids unsupported answers outside the course product domain.	3
29	Instructions	Tell the agent to state when an answer is not available in the product sheets.	Foundry Agent	Reduces hallucination and teaches safe fallback behavior.	3
30	Web app	Create a simple web app that sends user questions to the CloudXeus agent.	Azure App Service / Flask	Students see how the agent moves from portal testing into an application.	3

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Section 3 - Web App, Invoice Processing, Monitoring, and Release

Table 1 - CloudXeus Capstone Lab and Evidence Matrix (continued)

Rows continue from the previous page. Header row is repeated to preserve column meaning.

Step	Section	Requirement / Lab task	Azure service	Expected evidence	Pg
31	Web app	Display the answer and optional source references from the agent response.	Web application	The app makes grounded answers understandable to the end user.	4
32	Invoices	Upload student invoice PDFs to the student-invoices container for a separate processing flow.	Azure Blob Storage	Course product PDFs and invoice PDFs are stored separately for clarity and governance.	4
33	Invoices	Analyze invoice PDFs using Azure Content Understanding to extract invoice number, student name, date, course, and amount.	Azure Content Understanding	Structured invoice data can be reviewed, stored, or used by downstream workflows.	4
34	Monitoring	Enable Application Insights for the application and record agent calls, latency, and errors.	Azure Monitor	Students can troubleshoot the runtime behavior of the AI solution.	4
35	Monitoring	Add tracing to inspect retrieval, tool calls, model requests, and failures.	Application Insights	Operational visibility helps diagnose issues in production-like apps.	4
36	Reliability	Handle 429 throttling by reading retry guidance and applying exponential backoff.	Azure OpenAI / Foundry	Application retries gracefully instead of failing immediately during quota pressure.	4
37	DevOps	Store source code, prompts, configuration, and infrastructure scripts in GitHub.	GitHub	Changes are tracked and reviewable across the capstone project.	4
38	DevOps	Create a GitHub workflow to run tests and optional evaluation checks before deployment.	GitHub Actions	Students see how AI app changes can be validated before release.	4
39	Release	Deploy the web app and verify the end-to-end user experience.	Azure App Service	Users can ask questions and receive grounded CloudXeus course answers.	4
40	Review	Summarize the final architecture and explain how storage, search, Foundry, evaluations, and monitoring work together.	Capstone review	Students can explain the complete AI-103 capstone solution with confidence.	4

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